United States Department of the Interior

National Geospatial Technical Operations Center
Center of Excellence for Geospatial Information Science

December 10, 2025

To whom it may concern,

Please consider this letter of recommendation with Adam Camerer's application. I am a research scientist with the United States Geological Survey (USGS) Center of Excellence for Geospatial Information Science (CEGIS) in Rolla, Missouri where I have worked since 1998. CEGIS research develops and tests cutting edge technologies to support the production, distribution, display, and use of geospatial data that form the National Map for the United States. The National Map includes vector themes, such as data from the 3D National Hydrography Program, and raster themes, such as orthoimages from the National Agricultural Imagery Program (NAIP), and products furnished by the 3-D Elevation Program (3DEP), including high-resolution elevation data derived from light detection and ranging (lidar) and interferometric synthetic aperture radar (IfSAR) sensors. CEGIS also develops plans and workflows for science investigations and disseminates results through publications and presentations at various venues.

Adam worked with me from December 2022 through December 2024. During that time Adam helped develop tools and workflows to automate the extraction, representation, and evaluation of hydrographic features from digital elevation model (DEM) and remotely sensed data. Initially Adam improved his python skills and learned how to work with both raster and vector geospatial datasets, properly using associated spatial reference systems and available processing tools. In conjunction with his computer science courses, Adam became proficient at designing workflows to process hundreds of geospatial datasets using parallel computing techniques on available processing clusters.

Adam streamlined a parallel processing workflow to download and compile the various input layers used in our feature extraction models. Generally, our feature extraction methods involve training deep learning (DL) neural network models to predict hydrography from several input layers derived from IfSAR or lidar data, where DL predictions constrain the extraction of DEM-derived vector flow networks. The extracted hydrography helps update the United States national hydrography vector database, which stores all surface water features that are represented in USGS databases and topographic maps.

Adam developed numerous tools and workflows for various tasks using open source geospatial data analysis libraries and commercial packages to process both raster and vector data types. His tasks included vectorizing DL raster predictions, validating predictions, filtering reference data, and statistically summarizing and curating large databases of raster data. He created tools to generate and animate series of stream meander frequency plots, accompanied by synthesized musical tones. He established workflows for comparing feature densities of common datasets before and after generalization, allowing for a quantitative assessment of generalization workflows used in topographic mapping. He implemented plotting systems to visualize and evaluate test results effectively. Throughout these projects, Adam employed parallel processing techniques on CPU and GPU clusters to efficiently process large datasets.

Adam is a proficient software developer with a strong background in geospatial analysis and a solid understanding of deep learning techniques. He is skilled in Python and various geospatial data analysis modules, as well as open-source and commercial tools. His experience includes parallel processing across diverse cluster configurations and maintaining repositories such as Git. Additionally, he is familiar with Geographic Information Systems (GIS), including ArcGIS Pro and QGIS, for data visualization and review.

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During the two years working with Adam, Adam developed a strong understanding and capability for geospatial data analysis and machine learning. He demonstrated his ability to create efficient processing techniques in advanced computer science, topographic mapping, and environmental research. Adam has proven himself as a determined and hard worker, applying ingenuity to complete the tasks he undertakes.

It was a pleasure working with Adam; he was a productive programmer and instrumental in advancing our research efforts. Please let me know if you need any additional details.

Best regards,

LAWRENCE STANISLAWSKI

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